

Operating Systems (ST-1B)

1. Definition

An Operating System is system software that manages **hardware and software resources**.

Examples:

- **Windows**
 - **Linux**
 - **macOS**
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2. Functions of Operating System

1. Process management
 2. Memory management
 3. File system management
 4. Device management
 5. Security management
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3. Process Concept

A **process** is a program in execution.

Example:

Running browser, running game.

Process states:

- New
 - Ready
 - Running
 - Waiting
 - Terminated
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4. CPU Scheduling

CPU scheduling decides **which process runs first**.

Algorithms:

- FCFS (First Come First Serve)
- SJF (Shortest Job First)
- Round Robin